PnP-ADMM Tune-1: completed parameter search

PCM at 20% sampling with fixed NPL-default noise

Experiment results report, version 1.0 | 9 November 2026

Status: ADMM calibration complete (25/25); GS-PnP tuning and test set 45 pending.

1 Executive summary

We completed **25 independent Tune-1 calibration searches** for PnP-ADMM-DRUNet, with five randomly selected validation acquisitions from each of five PCM categories. All acquisitions used the original 20% Hadamard sampling masks and NPL-default noise observations. The hyperparameters were selected using held-out acquired measurements without access to ground-truth images or test image metrics.

The search evaluated **420 distinct parameter configurations**, each using four fixed SSDU folds, for **1,680 fold reconstructions**. All 25 selections are frozen. There were **zero failed candidates**. The mean of the 25 individually selected SSDU prediction errors was **0.004314767** (sample SD 0.000509679).

Scope: This report concerns *validation-only tuning*. The new configurations have not yet been evaluated on test set 45, because the agreed test runner requires the 25 GS-PnP selections to be frozen too. We therefore do not claim test PSNR, SSIM, HaarPSI or reconstruction superiority from these calibration losses.

2 Tuning algorithm

For one calibration acquisition $y = Ax + \varepsilon$ at 256×256 resolution and 20% multilevel Hadamard sampling ($J = 8$, coarse order $J_c = 6$), the 13,107 acquired measurements are split into four fixed SSDU folds. All 4,096 coarse rows remain in the fit set; the remaining 9,011 acquired fine rows provide held-out measurement subsets Λ_k.

For parameter vector $\theta = (\rho, \sigma)$, we minimise

$$L(\theta) = \frac{1}{4} \sum_{k=1}^{4} \frac{\|A_{\Lambda_k} \widehat{x}_\theta(A_{\Theta_k}, y_{\Theta_k}) - y_{\Lambda_k}\|_2^2}{|\Lambda_k|},$$

where the reconstruction only sees the complementary rows Θ_k. The denoiser weights and ADMM solver remain fixed.

Search in logarithmic coordinates with positive parameter bounds

$$\rho \in [0.03, 4.0], \qquad \sigma \in [0.015, 0.18].$$

Start at the geometric midpoint. For each coordinate, evaluate two interior points that divide its log interval into thirds, keeping the two-thirds interval favoured by the lower SSDU loss. Run **three contractions per coordinate and two coordinate passes with reversed order**, caching duplicate evaluations. Select the lowest *actually evaluated* finite SSDU risk.

This approach assumes *one-dimensional unimodality* of the objective while other parameters are fixed. That assumption has not been proved. It does not establish global optimality of the nonlinear joint PnP objective.

3 Step-by-step plan and execution

1. Freeze the validation calibration sample IDs, test45 sample IDs, RNG seed, archived acquisition SHA256 hashes, and parameter bounds. **Complete**.
2. Verify the old normalised PCM data-fidelity convention, four-fold measurement scoring, deterministic GPU behaviour, and numerical search implementation. **Complete**.
3. Run five independent ADMM parameter searches per category. Save candidate scores, original source checksums, fold diagnostics and numerical errors. **Complete, 25 of 25**.
4. Freeze each winner as an independent validation-only configuration. **Complete, 25 of 25**.

5. Run the separate GS-PnP calibration searches. **Pending, deliberately not started**.
6. When all 50 selections are frozen, reconstruct the fixed test set 45, nine images per category, using all acquired rows. **Pending**.
7. Compare with existing PCM methods using image PSNR, SSIM, HaarPSI, MSE, runtimes, and stored reconstruction arrays. **Pending**.

4 Experiment running and numerical checks

The ADMM search used **tuni's NVIDIA RTX 4070**, with deterministic CUDA/CuDNN settings and the original DRUNet weights. Each candidate requires four complete ADMM reconstructions, with the original maximum of 100 iterations and residual-based stopping. Candidate caching reduced the realised search from the maximum of 25 evaluations per calibration to a mean of **16.8**, with range 13–19. Each complete SSDU fold reconstruction took a mean of 1.253 seconds in the archived execution records, but timing varies with the chosen parameter and stopping behaviour.

Completed: 25 calibrations, 420 candidates, 1,680 fold reconstructions. No invalid candidates were recorded. The frozen selections and every scored fold remain on tuni. *The RTX 4070 was not used to evaluate test45 reconstruction quality in this phase.*

5 Results and deliverables

Five PCM category summaries

Category	Mean SSDU	SD SSDU	Median ρ	Median σ	Mean probes
CIGS	0.00372777	0.00009815	0.78297	0.05196	15.2
Organic	0.00435453	0.00013780	0.78297	0.07862	18.2
Perovskite	0.00404770	0.00045409	0.78297	0.05196	16.0
Reference	0.00462750	0.00052702	0.78297	0.05196	16.0
Silicone	0.00481633	0.00029584	0.37926	0.10362	18.6

Table 1: Each category contains five independently tuned validation measurements. Mean and sample SD describe selected held-out *measurement* MSE, not image reconstruction error.

All 25 independently selected ADMM configurations

Category	Validation ID	ρ	σ	SSDU MSE	Candidate evaluations
CIGS	0008	1.93756	0.05196	0.00366939	13
CIGS	0530	0.45461	0.07862	0.00363170	19
CIGS	0589	0.78297	0.05196	0.00367037	13
CIGS	0733	0.78297	0.04962	0.00384259	18
CIGS	0768	0.78297	0.05196	0.00382482	13
Organic	0158	0.78297	0.05965	0.00422714	18
Organic	0530	0.45461	0.08620	0.00446876	18
Organic	0589	0.78297	0.07862	0.00452695	18
Organic	0733	0.45461	0.08620	0.00422868	19
Organic	0768	0.78297	0.07862	0.00432111	18
Perovskite	0008	0.78297	0.05196	0.00446728	13
Perovskite	0045	0.65320	0.05965	0.00461581	18
Perovskite	0733	0.93853	0.05965	0.00372562	18
Perovskite	0768	0.78297	0.04962	0.00369531	18
Perovskite	0917	0.78297	0.05196	0.00373448	13
Reference	0008	0.93853	0.05196	0.00523090	13
Reference	0045	0.45461	0.07171	0.00474304	18
Reference	0158	0.65320	0.05196	0.00499911	13
Reference	0396	0.78297	0.07862	0.00410190	18

Category	Validation ID	ρ	σ	SSDU MSE	Candidate evaluations
Reference	0768	0.93853	0.04962	0.00406253	18
Silicone	0008	0.65320	0.12456	0.00526204	19
Silicone	0158	0.31640	0.07862	0.00483393	18
Silicone	0283	0.18371	0.12456	0.00479944	19
Silicone	0589	0.37926	0.10362	0.00443315	19
Silicone	0733	0.37926	0.10362	0.00475311	18

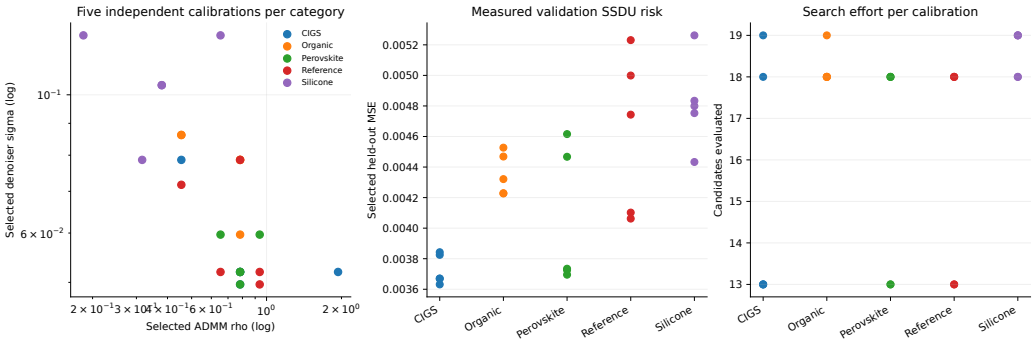


Figure 1: Selected regularisation/denoiser parameters, per-calibration SSDU prediction loss and candidate evaluation counts.

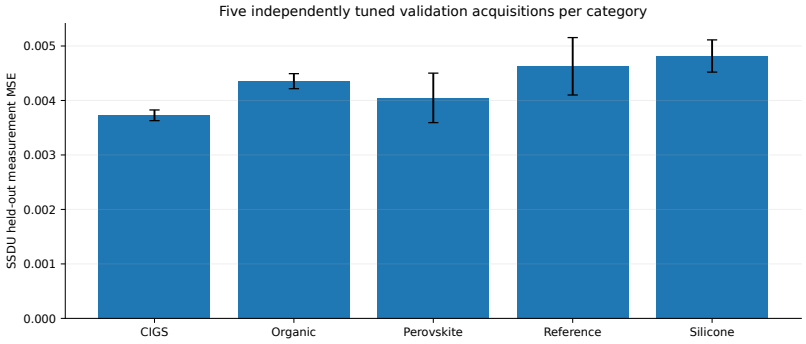


Figure 2: Within-category mean and sample SD of selected SSDU loss.

Reproducibility deliverables

The ZIP contains the full 25-selection JSON/CSV summary, all 420 candidate JSON score files with four SSDU-fold records each, all frozen parameter files and search traces, the fixed validation and test45 manifests, source scripts, operator/scoring provenance, and this editable LaTeX report with its figures and PDF. No test45 raw images or test quality metrics are included, because no test45 run has been performed.

6 Conclusion

The category-wise Tune-1 parameter search for PnP-ADMM is **complete and numerically successful**: all 25 independent validation-only selections were frozen, with no recorded candidate failure. The selected ρ and denoiser σ values vary across calibration acquisitions, so a single universal parameter pair is not established by this experiment.

A lower held-out measurement error does not necessarily imply higher image PSNR or preservation of thin lines. The next required study is transfer of each frozen configuration to its nine category-matched test45 images, followed by comparison with existing methods. GS-PnP calibration has been left untouched as requested.

Primary references. Yaman et al. (2020), SSDU measurement partitioning, https://doi.org/10.1002/mrm.28378.

Hurault, Leclaire and Papadakis (2022), gradient-step plug-and-play with backtracking, https://openreview.net/pdf?id=fPhKeld3Okz.